ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1 10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

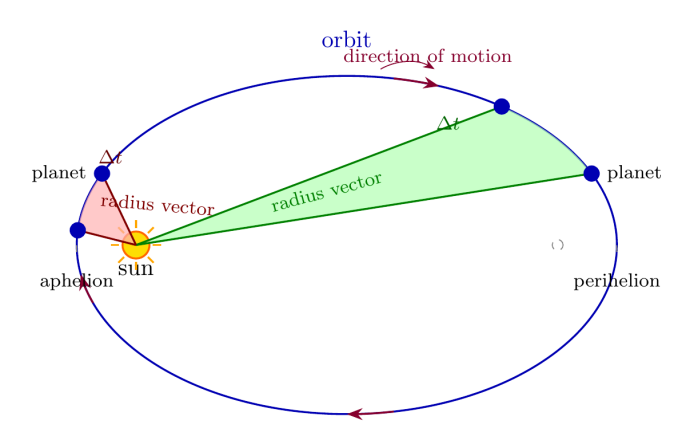

Fig. 1. An elliptical orbit diagram showing a planet orbiting the sun. The sun is located at one focus of the ellipse. A line from the planet to the sun sweeps out equal areas in equal times, illustrating Kepler's second law. The diagram should include the elliptical path, the sun at one focus, and a sector of the ellipse showing the area swept out over a time interval. Arrows should indicate the direction of the planet's motion and highlight that the area is larger when the planet is near aphelion and smaller when near perihelion, demonstrating the variation in speed.

Question 2 10.0 points

Examine the diagram below showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

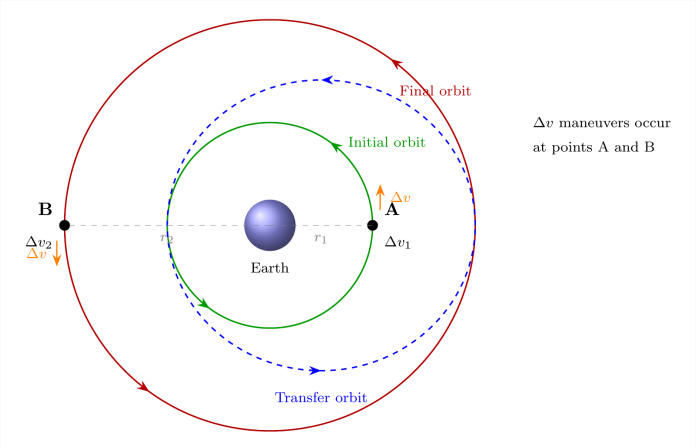

Fig. 2. MANDATORY LABELS — every item below MUST appear as visible text in the diagram: • Δv The diagram will FAIL automated review if any of these are missing. A Hohmann transfer orbit diagram showing two concentric circular orbits around Earth. The inner orbit is the initial orbit, and the outer orbit is the final orbit. The transfer orbit is an ellipse that is tangent to both circular orbits at the periapsis and apoapsis. Mark the points where velocity changes (Δv) occur: one at the periapsis of the transfer orbit (where it intersects the inner orbit) and one at the apoapsis (where it intersects the outer orbit). Include Earth at the center and label the orbits and Δv points. IMPORTANT: The diagram is missing required labels. The diagram is missing the required label "Δv". Add each missing label as visible text at the appropriate location in the diagram.

Question 3

10.0 points

Examine the diagram below illustrating the trajectory of a space probe executing a gravitational slingshot maneuver around a planet. Describe how this maneuver alters the probe's trajectory and velocity, and discuss the conservation laws that apply in this context.

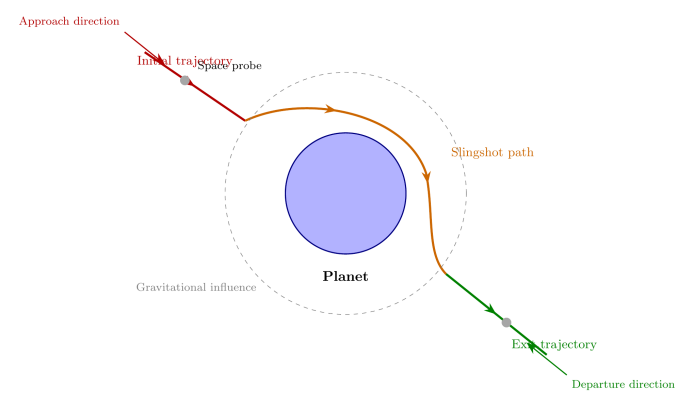

Fig. 3. A diagram showing the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. The diagram should include: a large planet at the center, the initial trajectory of the space probe approaching the planet, the curved path around the planet indicating the slingshot effect, and the exit trajectory of the probe. Arrows should indicate the direction of the probe's motion before and after the maneuver. Include labels for the planet, initial trajectory, slingshot path, and exit trajectory. IMPORTANT correction needed: Several labels in the top-left corner are. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: The diagram is missing required labels. The diagram has a typo in the "Initial trajectory. Add each missing label as visible text at the appropriate location in the diagram.

Question 4 10.0 points

Examine the diagram below illustrating the five Lagrange points (L1-L5) in the Earth-Sun system. Discuss the stability of each point and explain how satellites can utilize these positions.

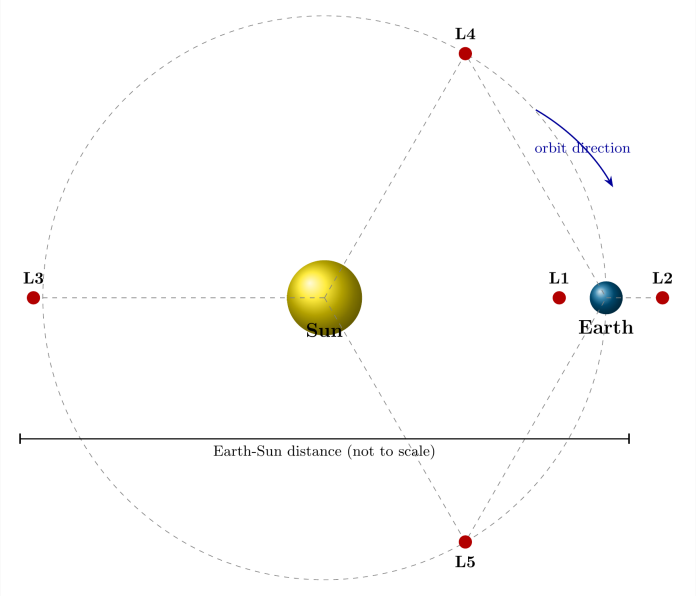

Fig. 4. A diagram showing the five Lagrange points (L1, L2, L3, L4, L5) in the Earth-Sun system. The Sun is on the left, and the Earth is on the right. L1 is located between the Earth and the Sun. L2 is on the opposite side of the Earth from the Sun. L3 is on the opposite side of the Sun from the Earth. L4 and L5 form an equilateral triangle with the Earth and the Sun, with L4 leading the Earth in its orbit and L5 trailing. Label each Lagrange point and show the Earth and Sun with appropriate distances and scales.

Question 5 10.0 points

Using the diagram below that details Mercury's orbit, explain the phenomenon of orbital precession and its causes, including contributions from both Newtonian mechanics and general relativity, with equations <eq_q_1> and <eq_q_2> resolved.

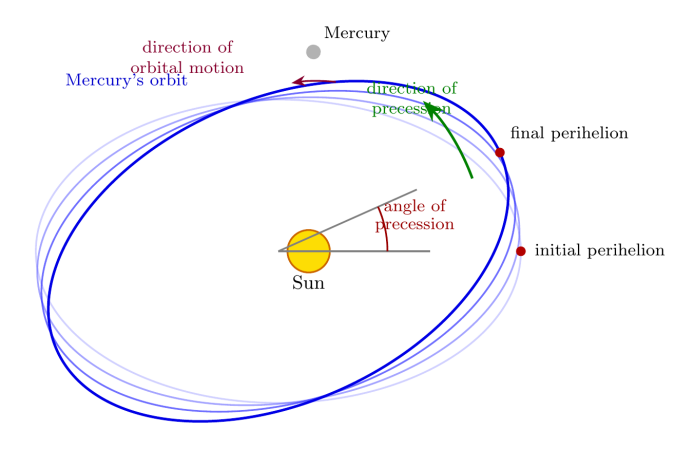

Fig. 5. Diagram of Mercury's orbit around the Sun, illustrating the phenomenon of orbital precession. The diagram should show an elliptical orbit with the Sun at one focus. Include an exaggerated depiction of the precession effect, where the orbit gradually shifts over time. Label the initial and final positions of the perihelion to highlight the precession. Indicate the direction of orbit and the angle of precession. Use arrows to show the direction of precession and the movement of Mercury along its orbit. IMPORTANT: The diagram is missing required labels. The diagram description requires an explicit label for the "angle. Add each missing label as visible text at the appropriate location in the diagram.